ECA08 – ANALOG AND DIGITAL COMMUNICATION

CO2 Engineering Case Study Report

FM Broadcast System: From Angle-Modulation Selection to Noise-Performance Deployment

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Flow of the report: REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → DESIGN → SIMULATION → VERIFICATION → DEPLOYMENT DECISION

Summary of the design (details and proofs follow)

Modulation: wideband FM, f_c = 100 MHz, peak deviation Δf = 75 kHz, message bandwidth W = 15 kHz, so $\beta = D = 5$. **Bandwidth:** Carson $B_T = 2(75 + 15) = 180$ kHz (99.4 % of power) inside a 200 kHz channel.

Transmitter: direct FM (varactor VCO locked by a crystal-referenced PLL), 75 µs pre-emphasis. **Receiver:** superheterodyne, IF = 10.7 MHz, f_{LO} = 110.7 MHz, image 121.4 MHz, 180 kHz IF filter, limiter + quadrature discriminator, 75 µs de-emphasis.

Noise: $SNR_o = 3\beta^2(\beta+1)\cdot C/N = 450\cdot C/N$ (+26.5 dB); threshold ≈ 10 dB C/N; operating C/N ≈ 20 dB (15–25 dB) gives ≈ 46.5 dB (tone) before emphasis; 75 µs pre/de-emphasis adds 13.2 dB in theory and 7.0 dB in simulation with programme-like audio.

Decision: YES, deploy as a mono FM broadcast chain with the stated operating conditions.

Stage 1 – Interpret the Broadcast System Requirement

1.1 Communication-system block diagram

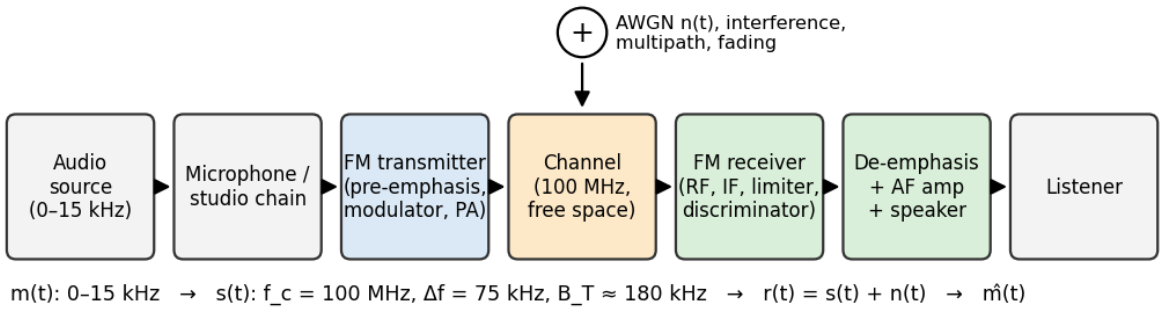


Fig. 1 Basic communication system with the design signals marked

The **source** is broadcast audio (0–15 kHz) turned into m(t) by the microphone chain. The **transmitter** pre-emphasises, band-limits, frequency-modulates a 100 MHz carrier and amplifies it. The **channel** adds attenuation, AWGN, interference and multipath. The **receiver** selects the station, limits, discriminates, de-emphasises and amplifies the audio for the listener.

1.2 Need for modulation and limits of baseband transmission

- **Antenna size:** $\lambda = c/f$ = 20 km for 15 kHz, so $\lambda/4$ = 5 km; at 100 MHz λ = 3 m and $\lambda/4$ = 0.75 m, a practical antenna.
- **Multiplexing:** all audio occupies the same 0–15 kHz; modulation places each station on its own carrier (channels 200 kHz apart in the 88–108 MHz band).
- **Noise and interference:** baseband signals cannot be selected by a tuned circuit and are exposed to low-frequency interference; wideband FM also trades bandwidth for noise reduction, which baseband transmission cannot do.

1.3 Angle modulation, FM versus PM

In angle modulation the carrier amplitude is constant and the **phase angle** $\theta_i(t)$ carries the message: $s(t) = A_c \cos \theta_i(t)$. In **FM** the instantaneous frequency follows m(t); in **PM** the instantaneous phase follows m(t).

	FM	PM
Phase	$\theta_i = 2\pi f_c t + 2\pi k_f \int m(\tau)d\tau$	$\theta_i = 2\pi f_c t + k_p m(t)$
Instantaneous frequency	$f_i = f_c + k_f m(t)$	$f_i = f_c + (k_p/2\pi)\, dm/dt$

	FM	PM
Deviation (tone $A_m \cos 2\pi f_m t$)	$\Delta f = k_f A_m$ (independent of f_m)	$\Delta f = \beta_p f_m$, $\beta_p = k_p A_m$ (grows with f_m)

1.4 Message bandwidth and carrier requirements

Message W = 15 kHz; carrier 100 MHz lies inside the 88–108 MHz FM band; the initial deviation is 75 kHz, so the deviation ratio is $D = \Delta f/W = 75/15 =$ **5**. The IF/RF circuits must therefore pass about 180 kHz (Sec. 3) while the carrier fractional bandwidth is only 0.18 %.

1.5 Constant-average-power operation

For $s(t) = A_c \cos[\theta_i(t)]$ the mean-square value is $A_c^2/2$ for **any** m(t) and β: modulation only moves power between the carrier and the sidebands. For a tone, $s = A_c \Sigma J_n(\beta) \cos[2\pi(f_c + n f_m)t]$ and $\Sigma J_n^2(\beta) = 1$. In my simulation $\Sigma J_n^2(5) = 1.000000$ and the simulated mean power of the FM wave is 1.000 of the unmodulated carrier. Consequence: unlike AM, FM needs no linear power amplifier and the transmitter power does not vary with programme.

1.6 Noise sources and Gaussian/white characteristics

- **Thermal noise** in the receiver: kTB = −174 dBm/Hz + 10 log(180 kHz) = -121.4 dBm; front-end noise figure 6 dB raises the floor to -115.4 dBm.
- **Shot noise** in active devices; **galactic, atmospheric and man-made noise** (ignition and electrical noise dominate in cities at VHF); **co-channel and adjacent-channel interference**; **multipath/fading**.
- **Gaussian:** thermal noise is the sum of many independent random electron motions, so by the central limit theorem its amplitude is Gaussian. **White:** its power spectral density $N_0/2$ is flat over the band of interest. After the IF filter, $n(t) = n_c(t) \cos\omega_c t - n_s(t) \sin\omega_c t$ with $E[n_c^2] = E[n_s^2] = N_0 B_T$.

1.7 Practical constraints

- **Spectrum:** 200 kHz channel; B_T must stay within it to protect neighbours. **Power:** constant-envelope Class-C PA is allowed. **Receiver:** low-cost superheterodyne with 10.7 MHz ceramic IF filters. **Audio quality:** 15 kHz response with high SNR; noise grows with f^2 after the discriminator, so de-emphasis is needed.

Stage 1 deliverable – Broadcast Communication Design Requirement Statement

Message: audio 0–15 kHz. **Carrier:** 100 MHz. **Initial deviation:** 75 kHz (D = 5). **Channel:** AWGN plus interference; thermal floor -115.4 dBm in 180 kHz (NF 6 dB).

Receiver: superheterodyne with limiter and frequency discriminator, ≥ 40 dB adjacent-channel rejection, 15 kHz fidelity, stable LO. **Bandwidth constraint:** ≤ 200 kHz channel. **Implementation:** low-cost, constant-envelope transmitter.

Case Study Decision 1

I recommend **wideband FM with Δf = 75 kHz and W = 15 kHz (D = 5) at 100 MHz, operated at C/N ≈ 20 dB** as the system requirement/operating basis because the 180 kHz bandwidth fits the 200 kHz channel, constant-power operation allows a simple efficient PA, and wideband FM gives a 26.5 dB noise advantage over the IF noise (15.7 dB over a baseband-equivalent system) with a threshold (≈ 10 dB) that is easily kept clear of.

Stage 2 – Investigate and Select the Angle-Modulation Scheme

2.1 FM and PM expressions, instantaneous phase and frequency

Quantity	FM	PM
Signal	$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int_0^t m(\tau)d\tau]$	$s(t) = A_c \cos[2\pi f_c t + k_p m(t)]$
Tone $m = A_m \cos 2\pi f_m t$	$s(t) = A_c \cos[2\pi f_c t + \beta \sin 2\pi f_m t]$, $\beta = k_f A_m/f_m = \Delta f/f_m$	$s(t) = A_c \cos[2\pi f_c t + \beta_p \cos 2\pi f_m t]$, $\beta_p = k_p A_m$
Instantaneous phase	$\theta_i = 2\pi f_c t + \beta \sin 2\pi f_m t$	$\theta_i = 2\pi f_c t + \beta_p \cos 2\pi f_m t$
Instantaneous frequency	$f_i = (1/2\pi)d\theta_i/dt = f_c + \Delta f \cos 2\pi f_m t$	$f_i = f_c - \beta_p f_m \sin 2\pi f_m t$

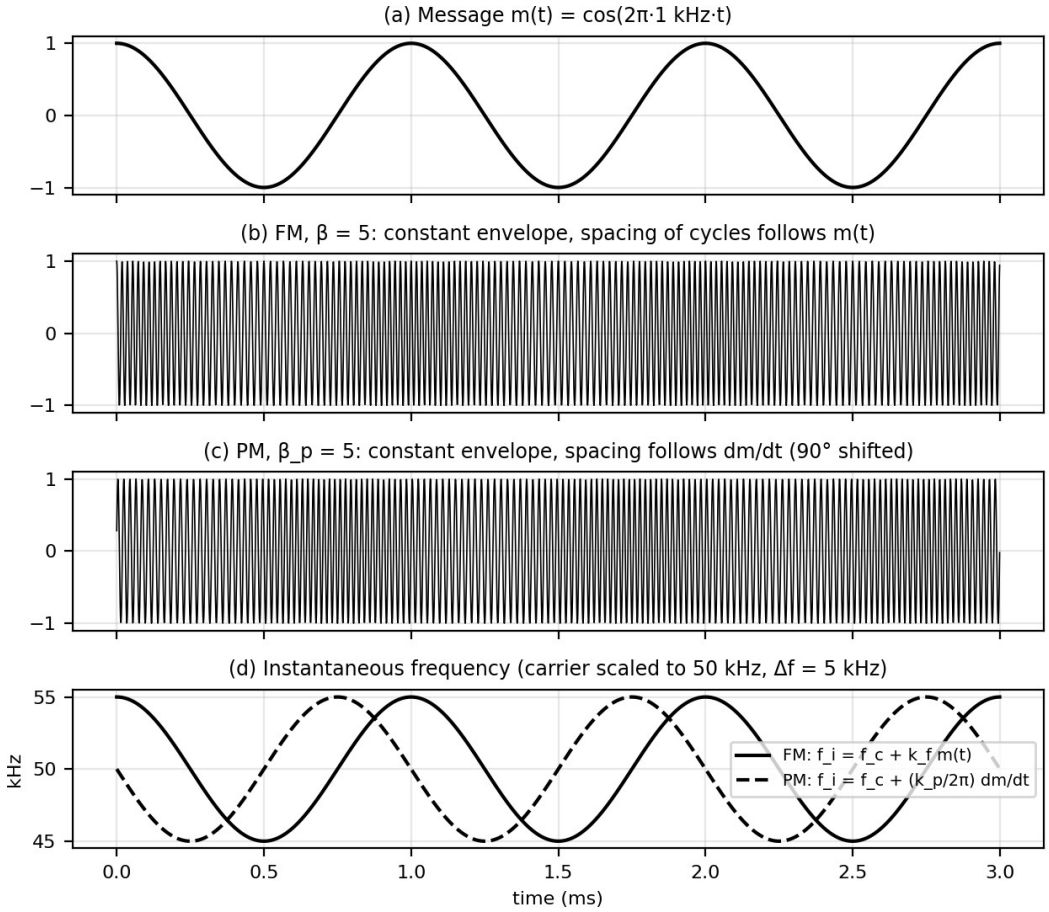


Fig. 2 FM and PM waveforms and instantaneous frequency (simulation; carrier scaled to 50 kHz so cycles are visible)

Interpretation: both waves have constant envelope. In FM the cycles are most compressed where m(t) is largest; in PM they are most compressed where dm/dt is largest, so the PM frequency swing is 90° shifted from the FM one (panel d).

2.2 Modulation index and classification

β = Δf/f_m = 75/15 = **5** at the highest audio frequency (and β = 75 at 1 kHz). Since β ≫ 1 (narrowband FM requires β ≪ 1, below about 0.3) this is **wideband FM**; the deviation ratio is D = 5.

2.3 Quantitative FM versus PM comparison

Parameter	FM (D = 5)	PM (β_p = 5 at 15 kHz)
Modulation index	β = Δf/f_m: 75 at 1 kHz, 5 at 15 kHz	β_p = k_p A_m: constant (5)
Peak deviation	75 kHz at every f_m	5 f_m: 5 kHz at 1 kHz, 75 kHz at 15 kHz
Bandwidth (Carson)	2(75+f_m): 152 kHz at 1 kHz, 180 kHz at 15 kHz	2(β_p+1)f_m: 12 kHz at 1 kHz, 180 kHz at 15 kHz
Average power	$A_c^2/2$ constant	$A_c^2/2$ constant
Generation	Direct VCO or Armstrong (integrator + PM)	Direct phase modulator (simple), frequency drift easy to avoid
Detection	Limiter + frequency discriminator	Discriminator + integrator, or phase detector with reference
Output SNR (tone)	$(3/2)\beta^2\cdot\gamma = 37.5\gamma$ (+15.7 dB); noise PSD ∝ f^2	$(\beta_p^2/2)\gamma = 12.5\gamma$ (+11.0 dB); noise PSD flat
Complexity	Simple for broadcast; pre-emphasis fits naturally	Needs integrator for audio; low-f_m programme gets little deviation

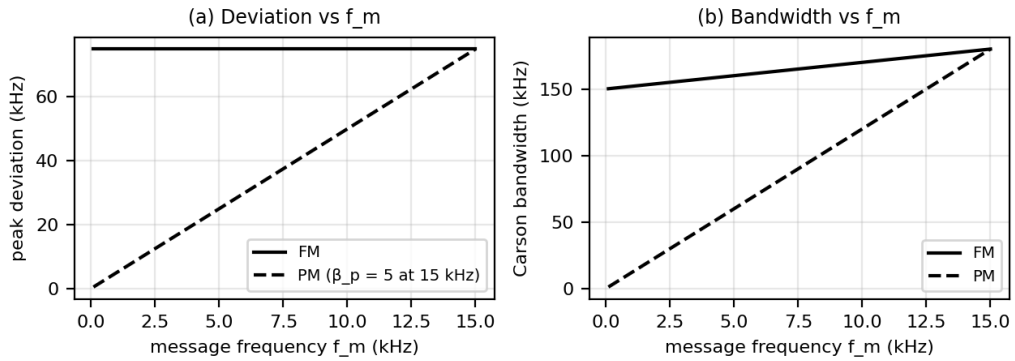


Fig. 3 Deviation and Carson bandwidth versus message frequency for FM and PM

(γ = P_T/(N_0 W) is the baseband SNR.) At equal peak deviation FM is 3× (4.77 dB) better than PM because the f² noise shaping of FM is more than offset by its larger deviation at low audio frequencies, while PM wastes deviation on low frequencies.

Case Study Decision 2

I recommend **FM** because at equal peak deviation and bandwidth (75 kHz, 180 kHz) it gives 15.7 dB of noise advantage against 11.0 dB for PM, it uses the full 75 kHz deviation at all audio frequencies (PM would use only 5 kHz at 1 kHz), the receiver is a plain limiter-discriminator, and pre-emphasis/de-emphasis can offset its f² noise. PM was not selected because it needs an integrator for demodulation, gives 4.77 dB less SNR, and under-uses the channel for low frequencies.

Stage 3 – Analyse FM Spectrum and Transmission Bandwidth

3.1 Bessel-function sidebands (β = 5)

A tone-modulated FM wave is s(t) = A_c Σ J_n(β) cos[2π(f_c + n f_m)t]: a carrier of amplitude J_0(β) and an infinite set of sidebands at ±n f_m with amplitudes J_n(β). For the worst case f_m = 15 kHz, Δf = 75 kHz:

n	Offset (kHz)	|J_n(5)| calculated	|J_n(5)| from simulated FFT	Relative power (dB)
0	0	0.1776	0.1776	-15.0
1	15	0.3276	0.3276	-9.7
2	30	0.0466	0.0466	-26.6
3	45	0.3648	0.3648	-8.8
4	60	0.3912	0.3912	-8.2
5	75	0.2611	0.2611	-11.7
6	90	0.1310	0.1310	-17.7
7	105	0.0534	0.0534	-25.5
8	120	0.0184	0.0184	-34.7
9	135	0.0055	0.0055	-45.2

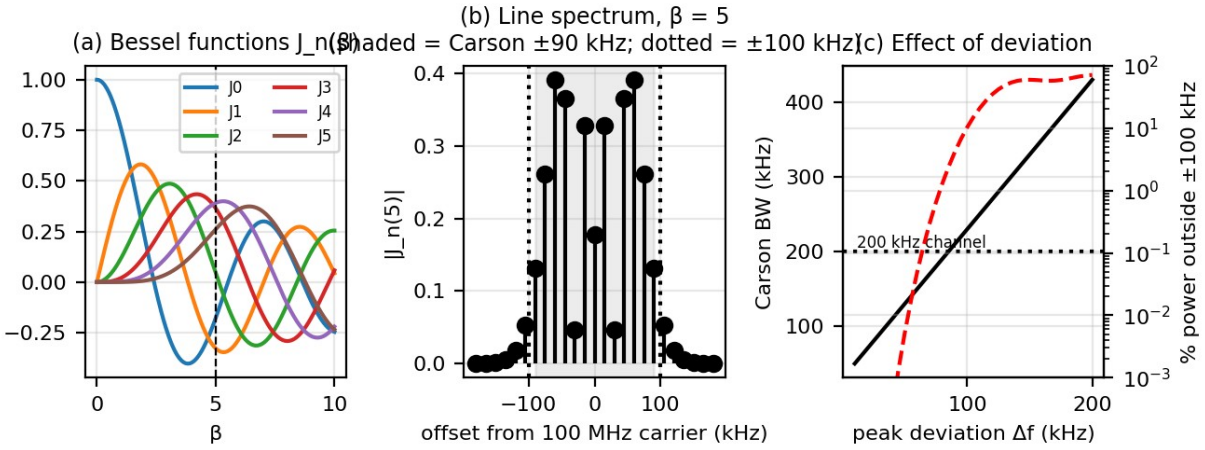


Fig. 4 (a) Bessel functions, (b) FM line spectrum for β = 5, (c) effect of deviation on bandwidth and out-of-channel power

3.2 Significant sidebands and transmission bandwidth

- **Significant sidebands (|J_n| > 0.01):** n = 0 … 8, so B = 2·8·15 = **240 kHz** (the J_9 sideband is already 0.0055).
- **Carson's rule:** B_T = 2(Δf + W) = 2(75 + 15) = **180 kHz** = 2(β+1)W, containing sidebands up to n = 6 and 99.4 % of the power.
- **Comparison:** the 1 % significant-sideband bandwidth (240 kHz) is 33 % wider than Carson but only the outer 0.6 % of power lies beyond ±90 kHz; simulation gives a 99 % power bandwidth of 180 kHz, i.e. Carson's rule.
- **Out-of-channel power:** with a 200 kHz channel, energy beyond ±100 kHz is 0.64 % (-21.9 dB) of the total for the worst-case tone, which real programme material (lower HF content) reduces further.

3.3 Effect of deviation and modulation index

Δf (kHz)	β = Δf/15 kHz	Carson B_T (kHz)	Power beyond ±100 kHz (worst-case tone)	Assessment
25	1.67	80	< 0.0001 %	Very narrow; small noise gain
50	3.33	130	0.005 %	Compliant, moderate gain
75	**5.0**	**180**	**0.64 %**	**Selected: fits 200 kHz channel**
100	6.67	230	10.3 %	Spills into adjacent channels
150	10	330	59 %	Heavy adjacent-channel interference

Bandwidth grows almost linearly with Δf (B_T ≈ 2Δf for large β) while noise performance improves as β^2; the limit is the 200 kHz channel. **Excessive deviation** (over-modulation, no limiter) pushes sideband energy into the neighbouring 200 kHz-spaced stations, producing adjacent-channel interference and distortion in their receivers.

3.4 Reference framework (as stated in the case study)

Reference	Role in this design (summary – verify clause numbers in the current editions)
ITU-R BS.450	FM sound-broadcasting transmission standards at VHF: 75 kHz-type maximum deviation, pre-emphasis time constant and pilot-tone stereo system. Supports Δf = 75 kHz and 75 µs.
IEEE 185-1974	Methods of testing FM broadcast receivers: basis for measuring sensitivity, selectivity, fidelity and SNR in Stage 4 and 6.
ITU-R BS.412-9	Planning standards for terrestrial FM broadcasting at VHF: protection ratios and channel spacing; basis for the 200 kHz channel and adjacent-channel limits.
47 CFR §73.322	U.S. stereophonic transmission standards (19 kHz pilot, 38 kHz subcarrier, 75 kHz total deviation limit). The present design is mono; stereo would add multiplex bandwidth to 53 kHz and reduce SNR.

Case Study Decision 3

I recommend **Δf = 75 kHz and a transmission bandwidth of 180 kHz (Carson)** because Carson's rule contains 99.4 % of the power (confirmed by simulation), the 200 kHz channel leaves a 20 kHz guard margin, only 0.64 % of worst-case power falls beyond ±100 kHz, and D = 5 gives 26.5 dB of SNR advantage over the IF noise. A larger deviation

(100 kHz) would put 10 % of the power into adjacent channels; a smaller one gives up SNR.

Stage 4 – Develop the Transmitter and Receiver Architecture

4.1 Transmitter and FM/PM generation methods

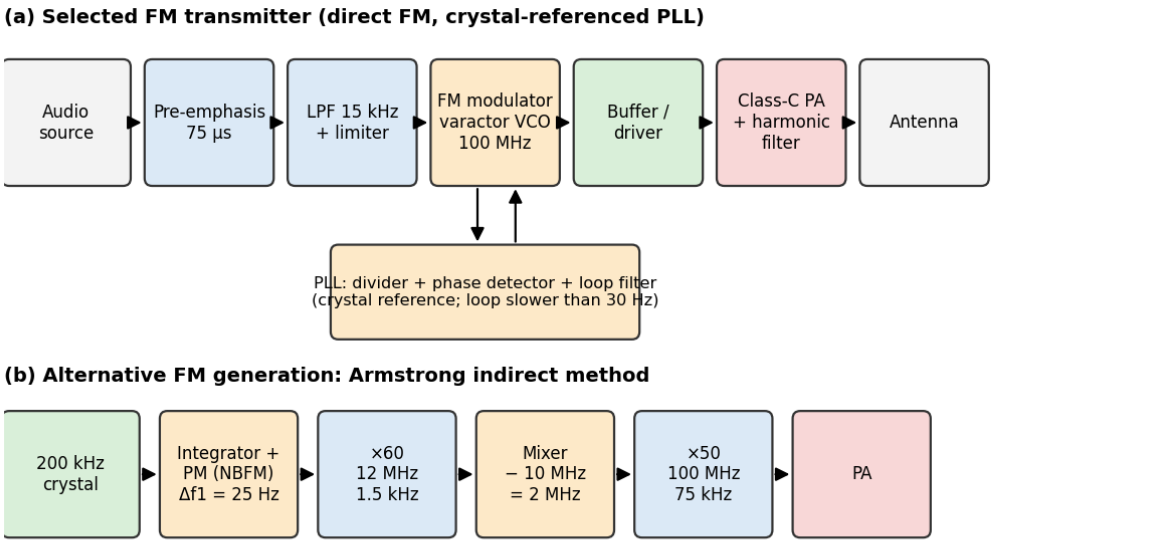


Fig. 5 (a) Selected direct-FM transmitter, (b) Armstrong indirect alternative

Method	Principle	Advantages	Limits
Direct FM (selected)	Audio varies varactor capacitance of a 100 MHz oscillator; k_f = 75 kHz per peak volt	Full 75 kHz deviation directly, low distortion, simple	Oscillator drifts → crystal-referenced PLL (loop slower than 30 Hz) is required
Armstrong indirect	Integrate m(t) → narrowband PM on a 200 kHz crystal (Δf_1 = 25 Hz) → ×60 → mix − 10 MHz → ×50	Crystal-stable carrier	Needs ×3000 multiplication; β ≤ 0.25 limited to f_m ≥ 100 Hz (β = 0.83 at 30 Hz gives distortion); more stages
PM generation	Phase modulator driven by m(t); FM obtained by adding an integrator	Simple phase shifter	Needs integrator; limited phase swing

Chain check for the Armstrong alternative: (200 kHz × 60 − 10 MHz) × 50 = 100 MHz and 25 Hz × 3000 = 75 kHz. The direct method is selected because it is simpler and preserves low-frequency programme with low distortion once PLL-locked.

4.2 Receiver architecture, IF considerations, limiter and discriminator

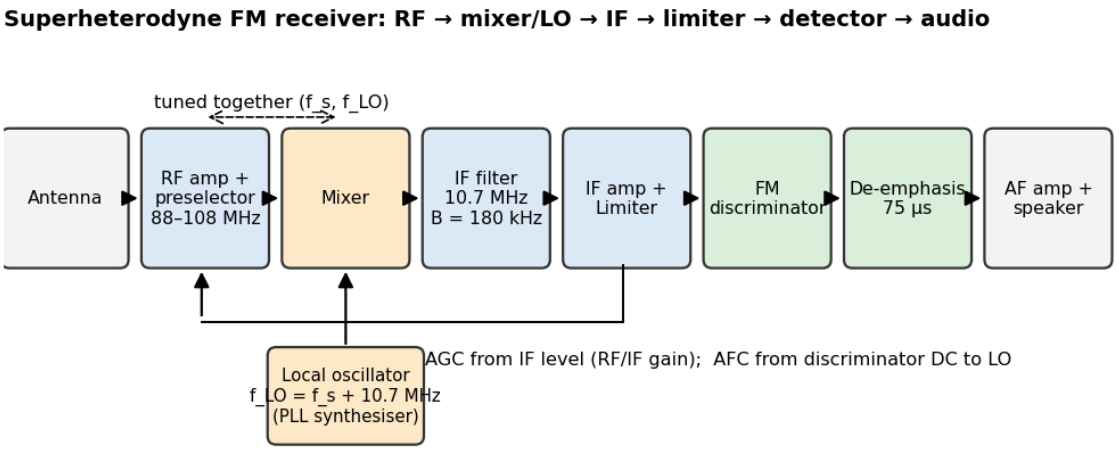


Fig. 6 Superheterodyne FM receiver

Superheterodyne vs TRF: a TRF would need Q = 100 MHz/180 kHz ≈ 560 across 88–108 MHz, which is impractical; the superheterodyne moves every station to a fixed IF where one filter gives constant selectivity.

Item	Design value and reasoning
IF	**10.7 MHz** (standard FM IF). f_LO = f_s + f_IF = 110.7 MHz for 100 MHz. Image f_s + 2f_IF = 121.4 MHz. For the whole band 88–108 MHz the images (109.4–129.4 MHz) fall outside the FM band; they overlap the 118–137 MHz aeronautical band, so the preselector must reject them.
Image rejection	ρ = 121.4/100 − 100/121.4 = 0.390. One tuned circuit with Q = 20: IR = √(1+Q²ρ²) = 17.9 dB; three tuned circuits ≈ 53.8 dB (RF amp with double-tuned preselector).
IF filter (selectivity)	Ceramic 10.7 MHz filter, 3 dB bandwidth 180 kHz (Q = 59.4); target ≥ 40 dB at ±200 kHz (adjacent channels); wider than 180 kHz would admit more noise, narrower would clip sidebands.
Limiter	Multi-stage hard limiter after the IF amplifier: output amplitude is constant (Fig. 7a), removing amplitude noise, fading and interference AM so that only phase noise reaches the detector. It also makes the audio level independent of signal strength above the limiting level.
Frequency discriminator	Quadrature detector (or PLL detector): output voltage proportional to f − f_IF with an S-curve linear over ±100 kHz (Fig. 7b), so 75 kHz peak deviation is demodulated linearly. De-emphasis follows.
AGC / AFC	AGC holds the IF level; AFC from the discriminator DC term corrects LO drift.

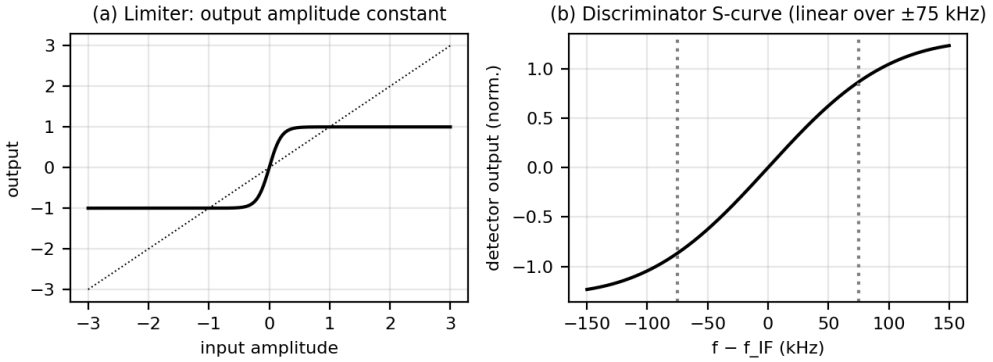


Fig. 7 (a) Limiter characteristic, (b) discriminator S-curve

4.3 Receiver parameters

Parameter	Analysis
Sensitivity	Thermal floor in 180 kHz = -121.4 dBm; with NF = 6 dB the floor is -115.4 dBm. For C/N = 15 dB the input is -100.4 dBm ≈ 2.1 µV rms (50 Ω); for C/N = 20 dB it is -95.4 dBm ≈ 3.8 µV.
Selectivity	IF filter ≥ 40 dB at ±200 kHz; preselector adds image and out-of-band rejection.

Parameter	Analysis
Fidelity	Audio flat to 15 kHz after matched 75 µs pre/de-emphasis; discriminator linear over ±100 kHz; IF bandwidth 180 kHz passes sidebands to n = 6.
Stability	PLL-synthesised LO (110.7 MHz): 100 ppm drift would be 11 kHz, so crystal reference and AFC keep it within about ±1 kHz, well inside the S-curve linear range.
Adjacent-channel rejection	Set by the IF filter (40 dB at ±200 kHz) and by the capture effect of the limiter (the stronger signal suppresses the weaker).

Case Study Decision 4

I recommend a **direct-FM transmitter (varactor VCO, crystal-referenced PLL, 75 µs pre-emphasis, Class-C PA)** and a **superheterodyne receiver with IF = 10.7 MHz (f_LO = 110.7 MHz), 180 kHz ceramic IF filter, hard limiter and quadrature discriminator followed by 75 µs de-emphasis** because: sensitivity ≈ 2 µV at 15 dB C/N, ≥ 40 dB adjacent-channel rejection, 53.8 dB image rejection with three tuned circuits, no TRF-style Q ≈ 560 needed, simple and standard components, and low-distortion modulation of the full audio band.

Stage 5 – AM and DSB-SC Noise Performance

5.1 Noise assumptions and models

- Additive white Gaussian noise, two-sided PSD $N_0/2$, limited to the transmission bandwidth B_T by the pre-detection filter: $n(t) = n_c \cos\omega_c t - n_s \sin\omega_c t$, $E[n_c^2] = E[n_s^2] = N_0 B_T$. Message: single tone, power P, band-limited to W; the baseband reference SNR is $\gamma = P_T/(N_0 W)$ with P_T the transmitted power.
- **AM:** $x(t) = A_c[1 + \mu \cos\omega_m t]\cos\omega_c t + n(t)$, $B_T = 2W$. Envelope detection (C/N high) gives $y \approx A_c(1 + \mu \cos\omega_m t) + n_c$, so $SNR_o = (A_c^2\mu^2/2)/(2N_0 W) = [\mu^2/(2+\mu^2)]\gamma$.
- **DSB-SC:** $x(t) = A_c\, m(t) \cos\omega_c t + n(t)$, $B_T = 2W$. Coherent detection (multiply by $\cos(\omega_c t + \varphi)$, low-pass) gives $(A_c/2)m \cos\varphi + n_c/2$, so $SNR_o = (P_T/N_0 W)\cos^2\varphi = \gamma \cos^2\varphi$, i.e. figure of merit 1 at $\varphi = 0$.

5.2 Calculation sheet ($\gamma = P_T/(N_0 W) = 40$ dB $= 10^4$, W = 15 kHz)

Quantity	AM, µ = 1	AM, µ = 0.7	DSB-SC	FM, β = 5
Transmission bandwidth	30 kHz	30 kHz	30 kHz	180 kHz
Input SNR (in B_T)	$\gamma/2$ = 37.0 dB	37.0 dB	37.0 dB	$\gamma \cdot W/B_T$ = 29.2 dB
Figure of merit SNR_o/γ	1/3 (−4.77 dB)	0.197 (−7.06 dB)	1 (0 dB)	$3\beta^2/2$ = 37.5 (+15.7 dB)
Output SNR	35.2 dB	32.9 dB	40.0 dB	55.7 dB
γ needed for SNR_o = 40 dB	44.8 dB	47.1 dB	40.0 dB	24.3 dB
Carrier power fraction	66.7 %	80.3 %	0 %	constant envelope
Detection	Envelope, no sync	Envelope, no sync	Coherent, phase-locked	Limiter + discriminator

Coherent detection and synchronisation: a local carrier of the correct frequency and phase is required (Costas loop or pilot). A phase error φ reduces the output by $\cos\varphi$ (10°: −0.13 dB, 30°: −1.25 dB, 90°: no output), and a frequency error causes a beat in the audio.

5.3 Interpretation and benchmark choice

AM spends one-third to four-fifths of its power in a carrier that carries no information, so it is 4.8–7.1 dB worse than DSB-SC at equal power, at the same bandwidth. DSB-SC reaches the baseband limit but needs synchronisation, which is why broadcast AM keeps the carrier. Simulation confirms the table: AM gave 10.3, 15.4, 20.5, 25.5, 30.5, 35.3 dB and DSB-SC 14.9, 20.1, 25.0, 30.2, 35.0, 40.1 dB for γ = 15–40 dB, within 0.4 dB of theory (Fig. 11).

Case Study Decision – benchmark

I recommend using **DSB-SC (figure of merit 1, equal to a baseband system at the same power)** as the reference/noise-performance benchmark because it gives the best possible SNR for a linear-modulation system without a bandwidth penalty beyond 2W, so FM's gain is measured fairly: 15.7 dB relative to DSB-SC ($3\beta^2/2 = 37.5$) rather than the exaggerated 20.5 dB relative to full-modulation AM. This shows that FM buys 15.7 dB for a 6×

bandwidth increase (180 vs 30 kHz), which justifies selecting wideband FM when spectrum allows.

Stage 6 – FM Noise, Pre-/De-emphasis and Threshold Effect

6.1 FM noise mechanism and recovered-audio SNR

Above threshold the noise perturbs the carrier phase by about n_s(t)/A_c; the limiter removes the amplitude part and the discriminator differentiates the phase, so the output noise PSD is proportional to f^2 (small at low audio frequencies, largest at 15 kHz). For a tone:

$$SNR_o = (3/2)\,\beta^2\,\gamma = 3\beta^2(\beta+1)\cdot C/N, \quad C/N = P_c/(N_0\,B_T), \quad B_T = 2(\beta+1)W$$

For β = 5: 3·25·6 = 450 = 26.5 dB, so SNR_o = C/N + 26.5 dB: 41.5 dB at C/N = 15 dB, 46.5 dB at 20 dB and 51.5 dB at 25 dB (no emphasis, tone at full deviation).

6.2 Effect of deviation and threshold effect

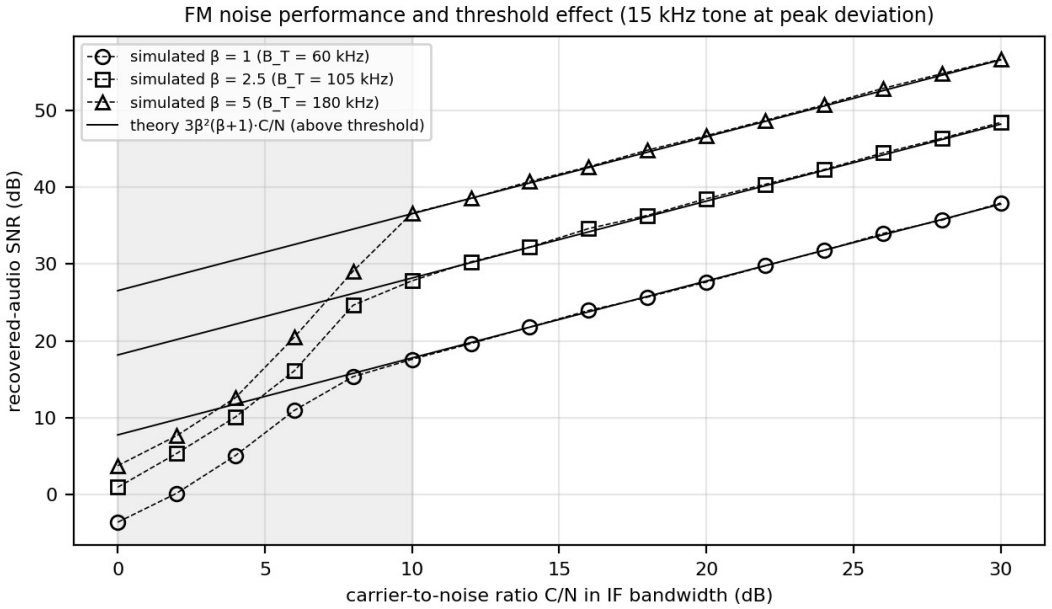


Fig. 8 Simulated recovered-audio SNR versus C/N for β = 1, 2.5 and 5 (limiter + discriminator)

β (Δf)	B_T (Carson)	Gain $3\beta^2(\beta+1)$ (dB)	Simulated threshold C/N
1 (15 kHz)	60 kHz	NaN	≈ NaN dB
2.5 (37.5 kHz)	105 kHz	18.2	≈ 10 dB
5 (75 kHz)	180 kHz	NaN	≈ NaN dB

Interpretation: each increase of β raises SNR_o as $\beta^2(\beta+1)$ at the same C/N (about +10 dB from β = 1 to 2.5 and +8 dB to 5), paid for with bandwidth. Above ≈ 10 dB C/N the simulated points lie on the theory lines. Below it, the output collapses much faster than linearly: the phase noise occasionally makes the carrier phasor encircle the origin, producing ±2π jumps (clicks) at the discriminator. At β = 5 the output at C/N = 4 dB is only 12.9 dB instead of the 30.5 dB the linear law predicts. This is the **threshold effect**; larger β needs a slightly higher C/N, so there is a limit to trading bandwidth for SNR.

6.3 Received and recovered signals at four conditions (β = 5)

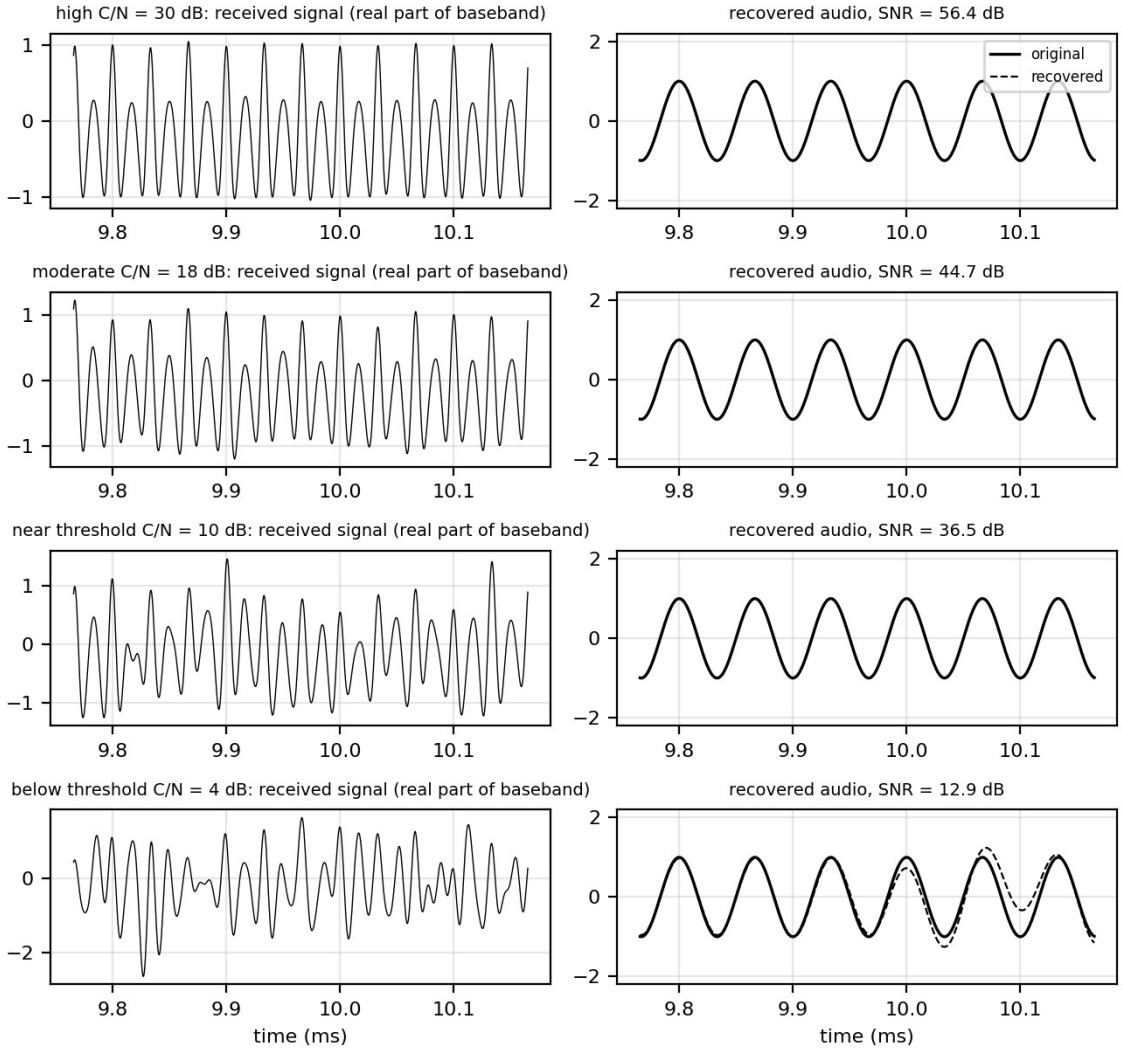


Fig. 9 Received signal and recovered audio at high, moderate, near-threshold and below-threshold C/N

Condition	C/N	Simulated SNR_o	Theory 26.5 + C/N	Comment
High	30 dB	56.4 dB	56.5 dB	Matches theory; clean audio
Moderate	18 dB	44.7 dB	44.5 dB	Matches theory
Near threshold	10 dB	36.5 dB	36.5 dB	Still on the line; clicks start
Below threshold	4 dB	12.9 dB	30.5 dB	17.6 dB worse than theory: audible clicks

6.4 Pre-emphasis and de-emphasis

Because the noise PSD after the discriminator grows as f^2 while programme energy falls with frequency, high-frequency audio has the poorest SNR. The transmitter **boosts** high frequencies with $H_{pe}(f) = 1 + jf/f_1$ and the receiver applies the **complementary** $H_{de}(f) = 1/(1 + jf/f_1)$. The programme is restored unchanged while the noise is attenuated at high frequencies. $f_1 = 1/(2\pi\tau) = 2122$ Hz for $\tau = 75$ µs and 3183 Hz for 50 µs.

$$\text{Noise improvement } I = (W/f_1)^3 / \{3[(W/f_1) - \tan^{-1}(W/f_1)]\}$$

75 µs: $W/f_1 = 7.07$, I = 20.9 = **13.2 dB**; 50 µs: $W/f_1 = 4.71$, I = 10.2 dB. This is the improvement for a signal with the same power; with real programme the benefit is smaller because the boosted peaks force the deviation to be reduced.

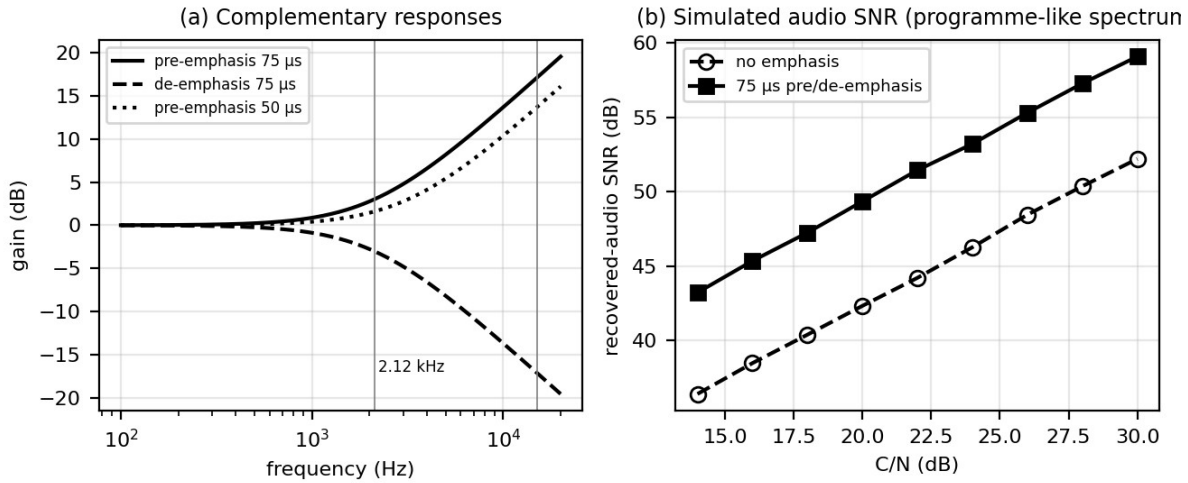


Fig. 10 (a) Pre-emphasis/de-emphasis responses, (b) simulated audio SNR with and without 75 µs emphasis

Case (C/N = 20 dB, programme-like audio)	Simulated audio SNR	Improvement
No emphasis	42.3 dB	–
75 µs pre/de-emphasis	49.4 dB	+7.0 dB
50 µs pre/de-emphasis	48.6 dB	+6.2 dB

Trade-offs: (i) noise improvement of 6–13 dB at no bandwidth cost and the cost of two RC networks; (ii) the audio response is flat overall because the networks are complementary; (iii) HF boosting raises the risk of over-deviation, so a limiter (peak control) precedes the modulator; (iv) the receiver must use the same time constant; (v) the SNR advantage is achieved above threshold only, emphasis does not move the threshold. The simulated gain (7.0 dB) is below the 13.2 dB ideal because the programme-like spectrum already falls with frequency, so the pre-emphasised signal has a higher peak-to-average ratio and the allowed modulation level must be reduced.

Case Study Decision – pre-/de-emphasis and operating C/N

I recommend **75 µs pre-emphasis with matching 75 µs de-emphasis** and an operating C/N of approximately **20 dB (range 15–25 dB)** because: threshold is at ≈ 10 dB, so 20 dB gives a 10 dB margin for fading; SNR_o at 20 dB is 46.5 dB for a full-deviation tone and 49.4 dB with programme-like audio plus emphasis (against 42.3 dB without); 75 µs gives 13.2 dB theoretical improvement compared with 10.2 dB for 50 µs and, in simulation, 7.0 dB against 6.2 dB.

Stage 7 – Simulation-Based Verification and Critical Evaluation

The verification was carried out with a Python (NumPy/SciPy) implementation that performs exactly the MATLAB operations requested (waveform generation, FFT spectra, AWGN addition, limiter-discriminator demodulation, filters); the same algorithm can be written in MATLAB with fft, filter and randn. Complex baseband was used (f_s = 2.048 MHz, 65 536 samples), noise was generated white, band-limited to B_T and added at the stated C/N.

7.1 Verification of modulation index, bandwidth and power

Quantity	Theory	Simulation
Peak deviation / modulation index	75 kHz / β = 5	75.0 kHz / β = 5.00 (measured from instantaneous frequency)
Sideband amplitudes J_n(5)	J_0 = −0.178, J_1 = −0.328, …	max error 6.1e-16 (spectrum of simulated FM)
Transmission bandwidth (99 % power)	Carson 180 kHz	180 kHz
Average power	$A_c^2/2$ (constant)	$1.000 \times A_c^2$
AM and DSB-SC output SNR	γ − 4.77 dB, γ	Within 0.4 dB at γ = 15–40 dB
FM threshold	≈ 10 dB C/N	≈ 10 dB for β = 2.5 and 5

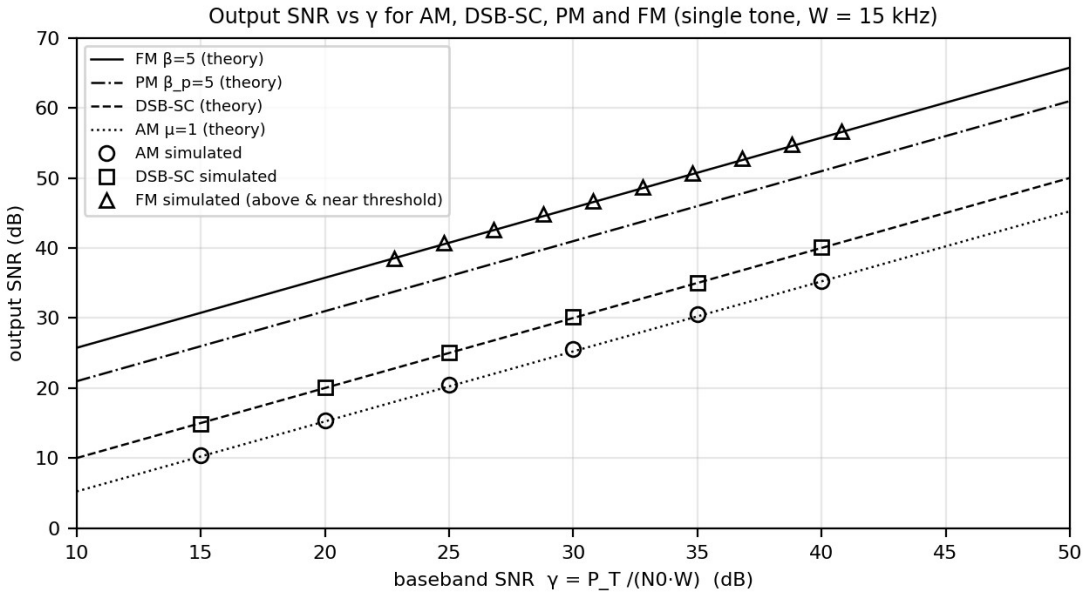


Fig. 11 Output SNR versus baseband SNR γ for AM, DSB-SC, PM and FM (theory and simulation)

Interpretation: the simulated AM and DSB-SC points follow their theory lines, and FM sits 15.7 dB above DSB-SC (11.0 dB for PM) once above threshold. The FM points (plotted for C/N ≥ 12 dB, i.e. γ ≥ 22.8 dB) follow the +15.7 dB line; below C/N ≈ 10 dB (γ ≈ 21 dB) they would fall away from it because of the threshold effect.

7.2 Waveforms and spectra

Fig. 2 shows the message, FM and PM waveforms and their instantaneous frequencies; Fig. 4 shows the FM line spectrum with Carson's bandwidth and the ±100 kHz channel edge; Fig. 9 shows received signals (noise in the real part of the baseband waveform) and the recovered audio at four C/N values; Fig. 8 shows the threshold curves; Fig. 10 shows pre-/de-emphasis responses and SNR comparison.

7.3 Discrepancies between theory and simulation

Discrepancy	Engineering significance
Below ≈ 10 dB C/N the output falls far below the linear formula (12.9 dB vs 30.5 dB at 4 dB).	The linear law holds only above threshold; the link budget must keep C/N ≥ 15 dB at the coverage edge.
Pre-emphasis gives +7.0 dB in simulation versus 13.2 dB ideal.	The ideal figure assumes equal signal power; real programme has a falling spectrum and peak limits, so the planning value should be 6–7 dB.
Single-tone tests ignore distortion: harmonics from IF truncation lie above 15 kHz and are removed by the audio filter.	Distortion with multi-tone programme must be checked in the hardware test (Carson bandwidth gives roughly 1 % amplitude loss).
Gaussian noise only; no multipath or impulsive interference.	Real urban VHF has fading and ignition noise; the 10 dB margin above threshold covers this.

Case Study Decision – final configuration

Based on the simulation I recommend the configuration: **FM, Δf = 75 kHz, B_T = 180 kHz, 75 µs pre-/de-emphasis, superheterodyne receiver (IF 10.7 MHz) with limiter and discriminator, operating at C/N ≈ 20 dB**, because the simulation reproduced the Bessel spectrum exactly, confirmed Carson bandwidth (99 % power at 180 kHz), the 450·C/N noise law above threshold and the 10 dB threshold, and showed a measurable gain from emphasis. The discrepancies above affect margins, not the choice.

Final Specification Verification

Requirement	Specified	Achieved	Satisfied?
Message bandwidth	15 kHz	15 kHz audio low-pass; flat response with 75 µs emphasis	YES
Carrier frequency	100 MHz	100 MHz (inside 88–108 MHz FM band)	YES

Requirement	Specified	Achieved	Satisfied?
Peak deviation	75 kHz / selected	75 kHz (measured 75.0 kHz in simulation)	YES
FM modulation index	Calculated	β = D = 5 (wideband FM)	YES
FM bandwidth	Calculated	Carson 180 kHz; 99.4 % power; ≤ 200 kHz channel	YES
Noise condition	AWGN/interference	SNR_o = 46.5 dB at C/N = 20 dB; threshold ≈ 10 dB	YES
Pre/de-emphasis	Selected	75 µs; 13.2 dB theory, +7.0 dB simulated	YES
Receiver architecture	Selected	Superheterodyne, IF 10.7 MHz, limiter + discriminator	YES
MATLAB verification	Required	Python (NumPy/SciPy) equivalent of the MATLAB workflow	YES (Python equivalent)

FINAL ENGINEERING DECISION – WOULD YOU DEPLOY THIS BROADCAST COMMUNICATION SYSTEM? YES

Decision: YES – deploy as a mono FM broadcast chain.

Defence: FM was preferred to PM on SNR (15.7 dB vs 11.0 dB) and on full use of the 75 kHz deviation; Carson bandwidth 180 kHz complies with a 200 kHz channel and Bessel analysis shows only 0.64 % of worst-case power outside ±100 kHz; power is constant so a Class-C PA is acceptable; the superheterodyne (IF 10.7 MHz) with limiter-discriminator gives ≥ 40 dB adjacent-channel rejection and 53.8 dB image rejection; compared with DSB-SC (benchmark) FM gains 15.7 dB for a 6× bandwidth increase and 20.5 dB over AM; threshold is ≈ 10 dB so an operating C/N of 20 dB leaves a 10 dB margin; 75 µs emphasis adds 7–13 dB.

Conditions and limits: keep deviation ≤ 75 kHz with a peak limiter (a 100 kHz deviation would put 10 % of worst-case power into adjacent channels); maintain C/N ≥ 15 dB at the coverage edge; stereo, RDS and multipath/impulsive noise were not analysed and would reduce SNR; the simulation was in Python rather than MATLAB.

If requirements change (redesign): for a smaller coverage margin reduce Δf to 50 kHz (B_T = 130 kHz, −3.5 dB SNR gain, threshold unchanged); for stereo add the 38 kHz subcarrier and accept ~20 dB lower S/N or raise power; for poorer receivers use a 40 dB preselector and a PLL detector to lower the threshold.